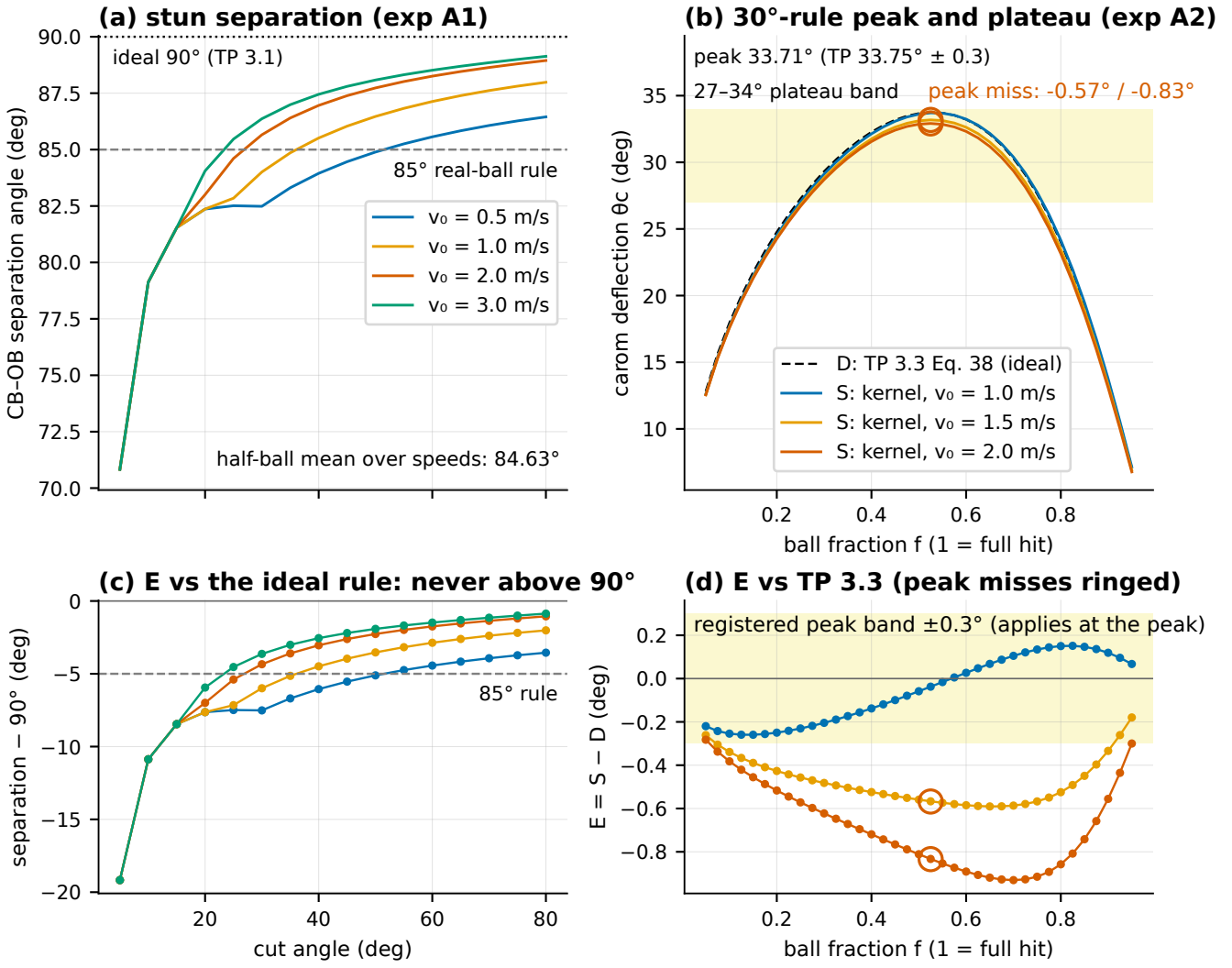
(a) stun separation (exp A1)
ideal 90° (TP 3.1)
85° real-ball rule
half-ball mean over speeds: 84.63°
v₀ = 0.5 m/s
v₀ = 1.0 m/s
v₀ = 2.0 m/s
v₀ = 3.0 m/s
CB–OB separation angle (deg)
90.0
87.5
85.0
82.5
80.0
77.5
75.0
72.5
70.0
(b) 30°-rule peak and plateau (exp A2)
peak 33.71° (TP 33.75° ± 0.3)
27–34° plateau band
peak miss: -0.57° / -0.83°
D: TP 3.3 Eq. 38 (ideal)
S: kernel, v₀ = 1.0 m/s
S: kernel, v₀ = 1.5 m/s
S: kernel, v₀ = 2.0 m/s
carom deflection θc (deg)
35
30
25
20
15
10
0.2
0.4
0.6
0.8
ball fraction f (1 = full hit)
(c) E vs the ideal rule: never above 90°
85° rule
separation − 90° (deg)
0
−5
−10
−15
−20
20
40
60
80
cut angle (deg)
(d) E vs TP 3.3 (peak misses ringed)
registered peak band ±0.3° (applies at the peak)
E = S − D (deg)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0.2
0.4
0.6
0.8
ball fraction f (1 = full hit)